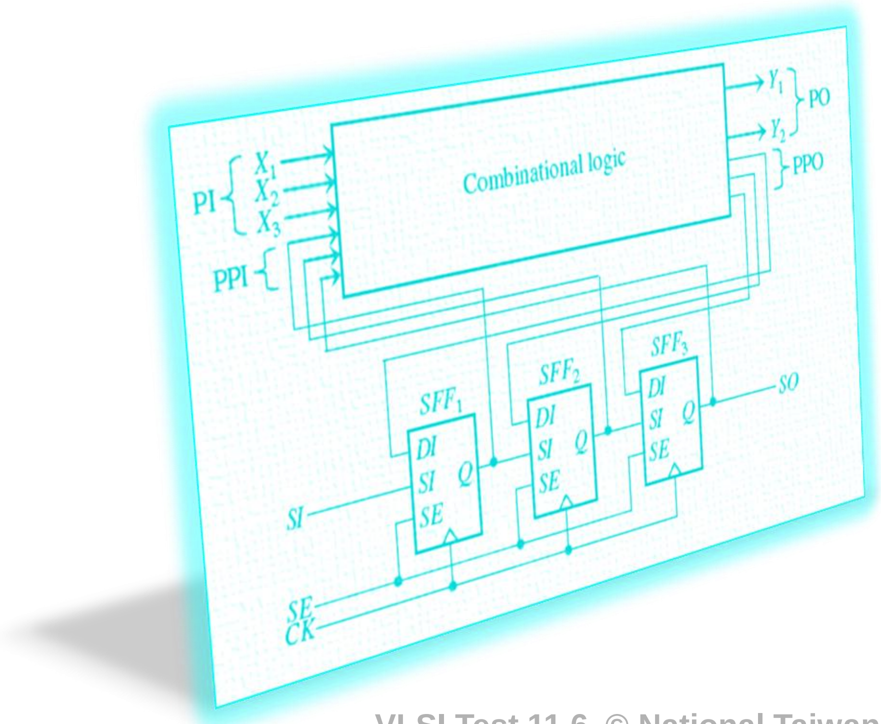
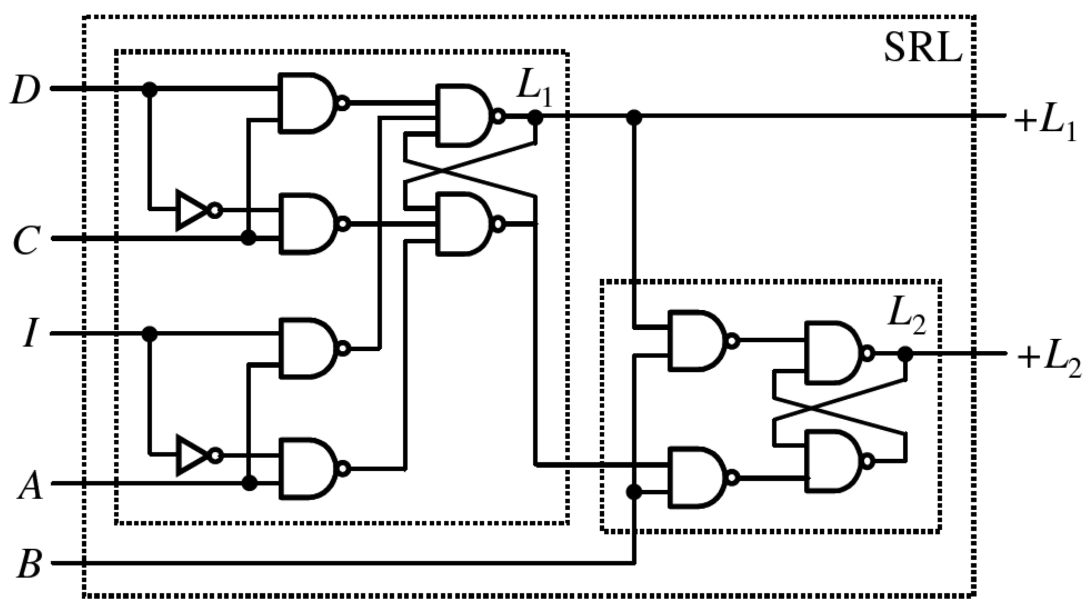
DFT - Part 1

- Introduction
- **Internal Scan**
  - FF-based
    - MUXed-D scan (1973, Stanford)
    - Clocked scan (1968, NEC)
    - Other scan
  - **Latch-based**
    - **LSSD (1977, IBM)**
- Scan Design Flow
- Issues and Solutions
- Conclusion



Level Sensitive Scan Design, LSSD

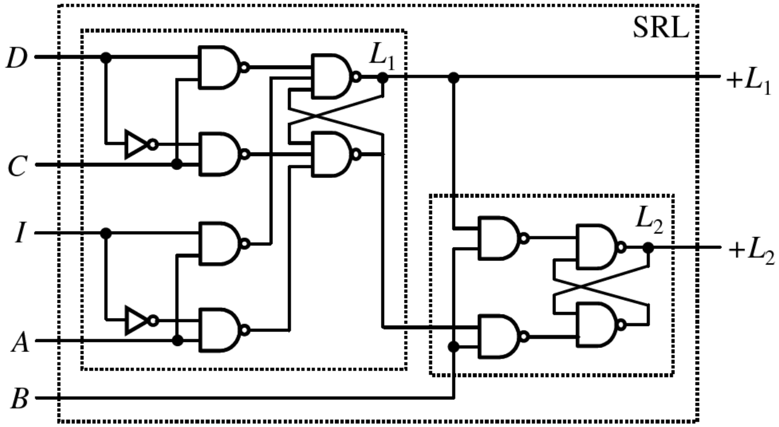
- Invented by IBM [Eichelberger 77]
- Master-slave latch
  - **Master L_1**: 2-port latch, AC clocks
  - **Slave L_2**: 1-port latch, B clock
  - 2 inputs: Data input (*D*); Scan input (*I*)
  - 2 outputs: *+L1*, *+L2*



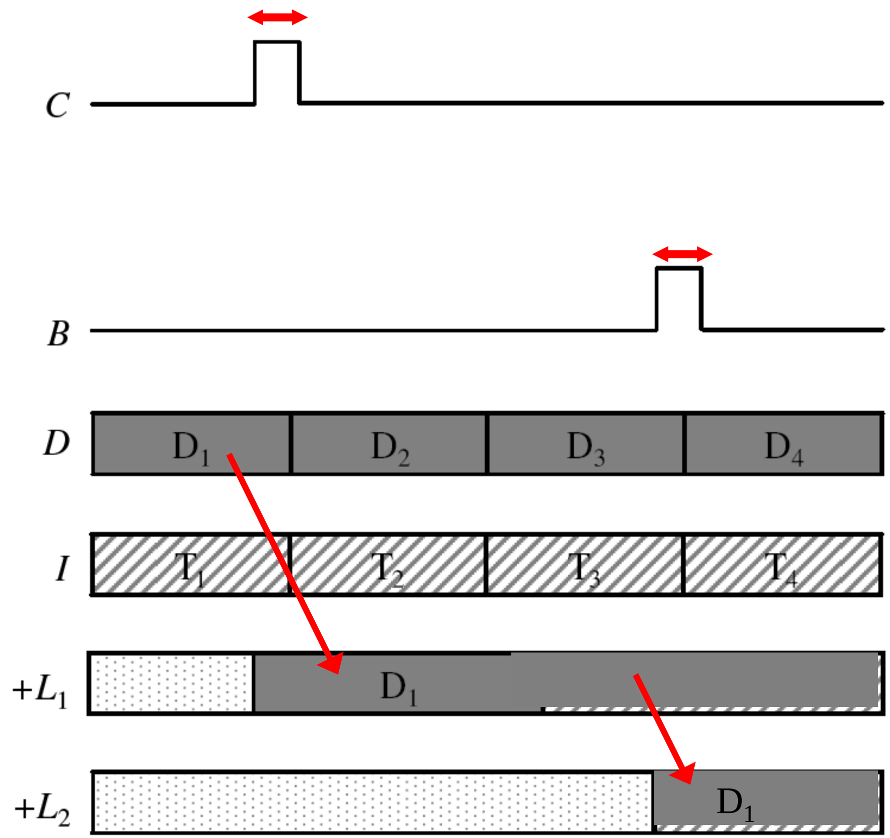
LSSD scan latch is called: *Shift Register Latch (SRL)*

Normal Mode Operation

- **Clocking sequence:**
  - **C clock B clock: D → +L_1 → +L_2**

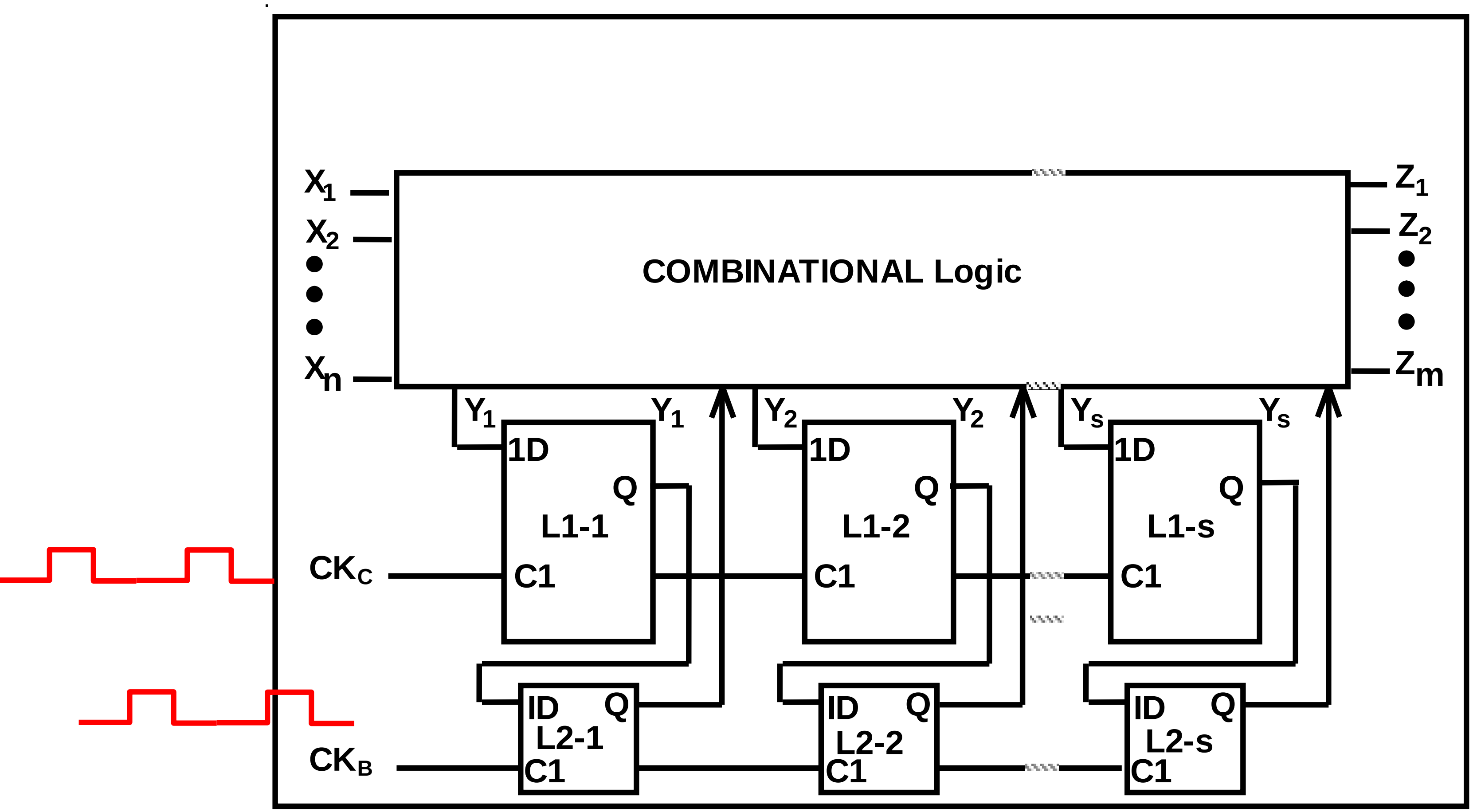


(WWW Fig. 2.12)



Double Latch-based Design (w/o LSSD)

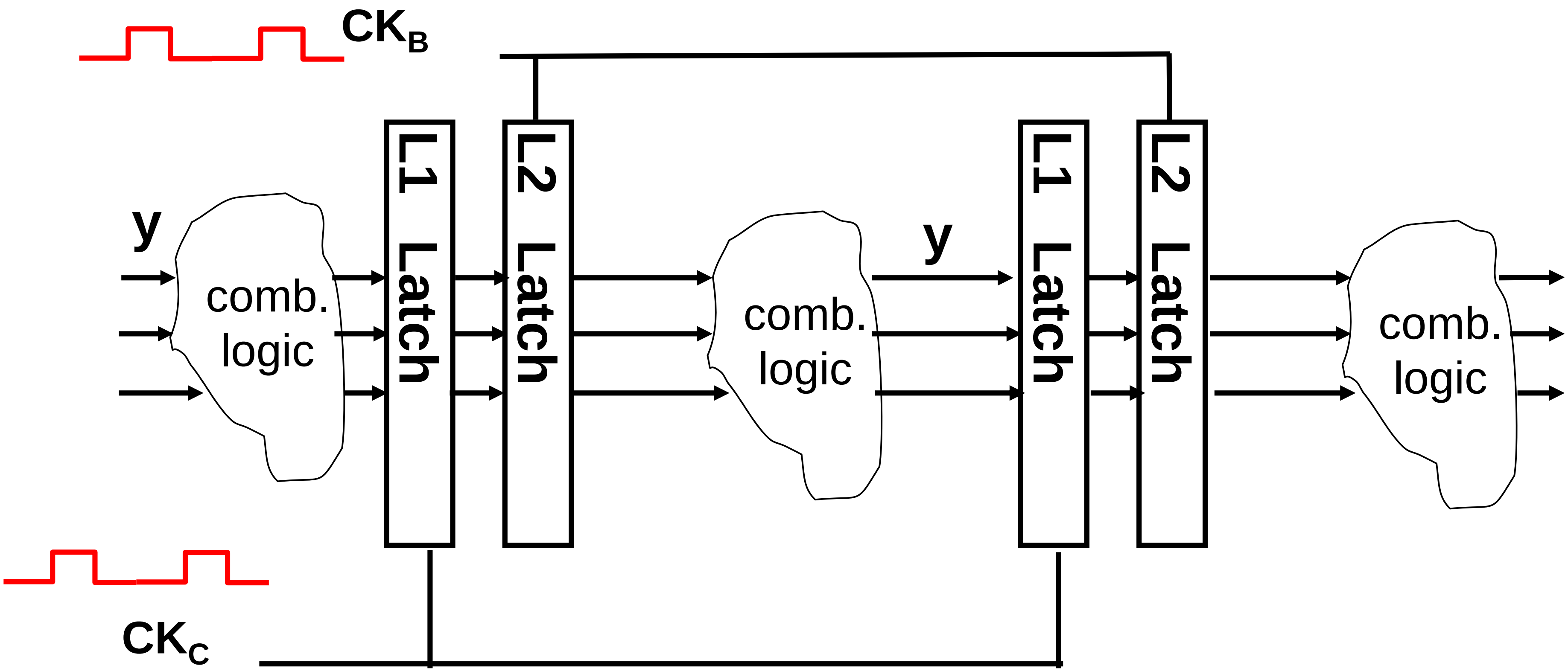
- Normal operation mode: CK_C CK_B, CK_C CK_B …
  - Clock B and C non-overlapping



(Courtesy of Prof. McCluskey, Stanford)

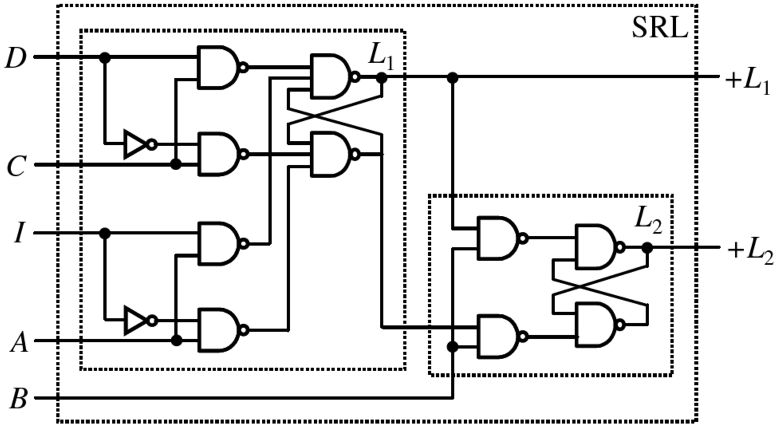
Another Point of View

- 2 latches per stage: L1 and L2
- Normal mode: CK_C CK_B, CK_C CK_B …
  - non-overlapping clocks, avoid clock skew

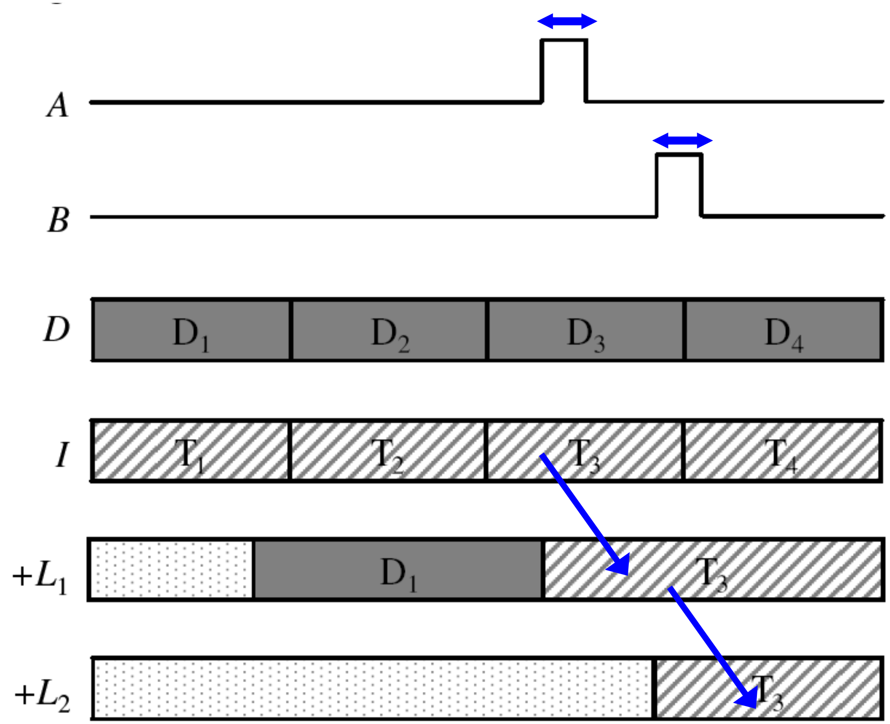


Shift

- **Clocking sequence:**
  - **A clock B clock: I → +L_1 → +L_2**

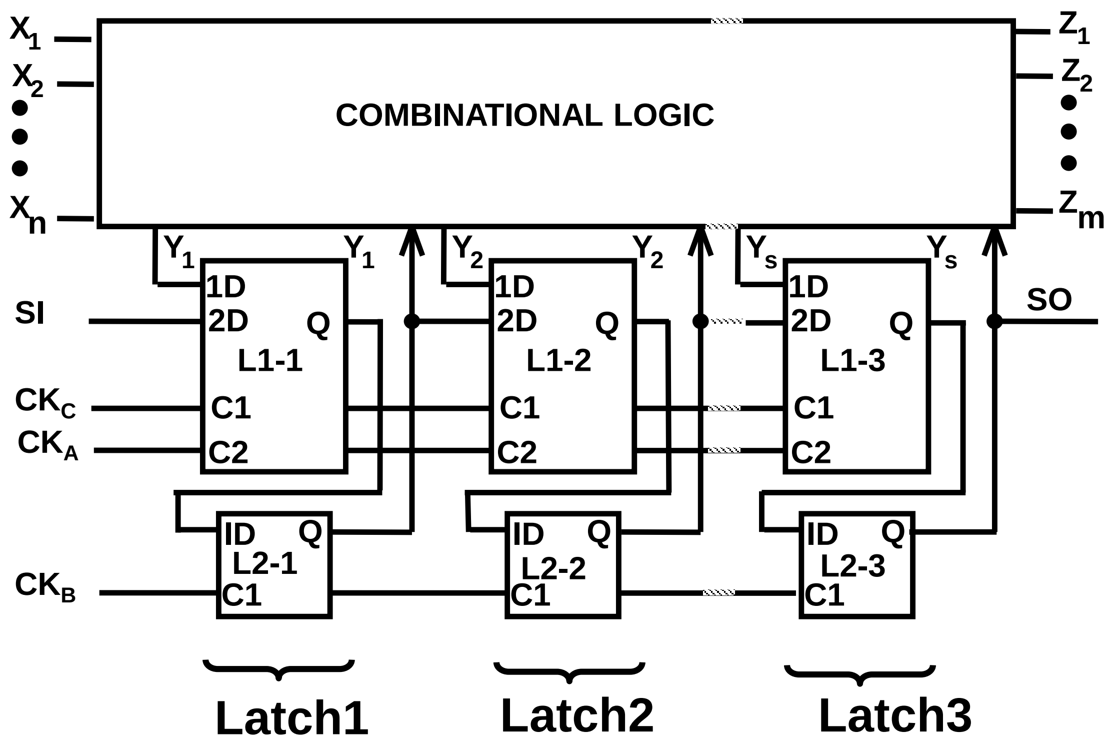


(WWW Fig. 2.12)



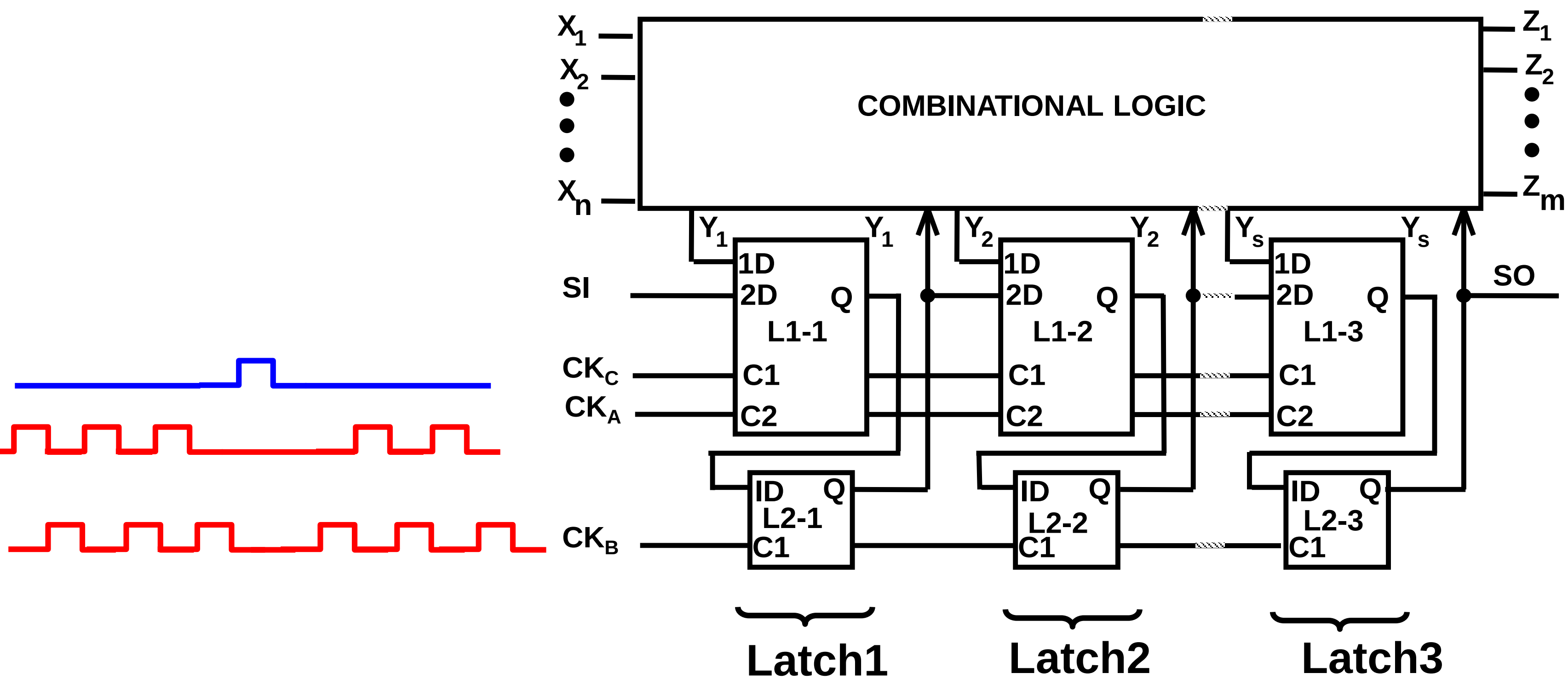
Double Latch-based Design (w/ LSSD)

- Replace L_1 1-port latch by 2-port latch
- Add scan input (SI), scan output (SO)
- Add A clock



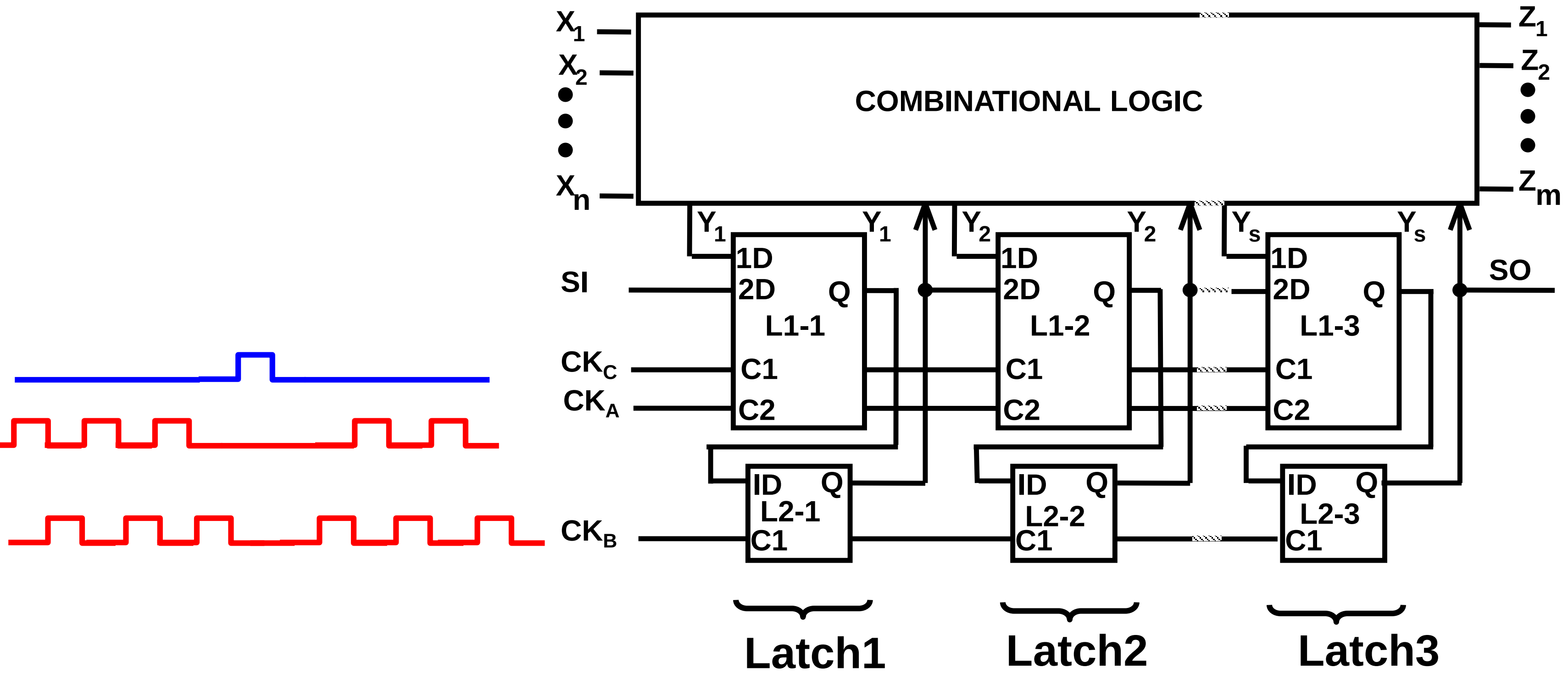
Test Mode Operation (1)

- One-pattern test
  - Shift: CK_A CK_B, CK_A CK_B , CK_A CK_B
  - Capture: CK_C
  - Shift: CK_B CK_A, CK_B CK_A, CK_B



Test Mode Operation (2)

- One-pattern test
  - Shift: CK_A CK_B, CK_A CK_B , CK_A CK_B
  - Capture: CK_C
  - Shift: CK_B CK_A, CK_B CK_A, CK_B



Summary of Three Internal Scan

- **LSSD**
  - ☺ **No clock skew**
  - **popular for latch-based design**

- **Muxed D-Scan**
  - ☺ **Very popular for FF-based design**
  - ☹ **Speed degradation: one MUX delay (about 2 gate delay)**
  - **Adopted by most ASIC designs**
- **Clock Scan**
  - ☺ **Little speed degradation**
  - ☹ **Need extra clock routing for SCK**
  - **Adopted by advanced designs**

FFT

- Q: how to apply 2-pattern test for LSSD?